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Splunk SIEM Security Monitoring & Threat Detection Lab

Project Overview

This project demonstrates the use of Splunk SIEM for security monitoring, log analysis, and threat detection. A controlled lab environment was created using a prebuilt dataset to simulate real-world attack scenarios. I analyzed authentication logs, investigated suspicious activity, and configured detection mechanisms to identify potential security threats.

Lab Setup & Data Ingestion

To simulate a realistic security environment, I installed Splunk Enterprise and ingested the BOTS v3 dataset. This dataset contains logs representing normal and malicious activity across multiple systems. The data was indexed and made searchable within Splunk to support analysis and detection use cases.

Tools & Technologies

- Splunk Enterprise
- SPL (Search Processing Language)
- Windows Event Logs
- Data Visualization (Dashboards)

Objectives

- Analyze authentication logs for suspicious behavior
- Detect brute-force login attempts
- Identify abnormal file modification activity
- Create dashboards to visualize security events
- Configure alerts for automated threat detection

Screenshots & Analysis

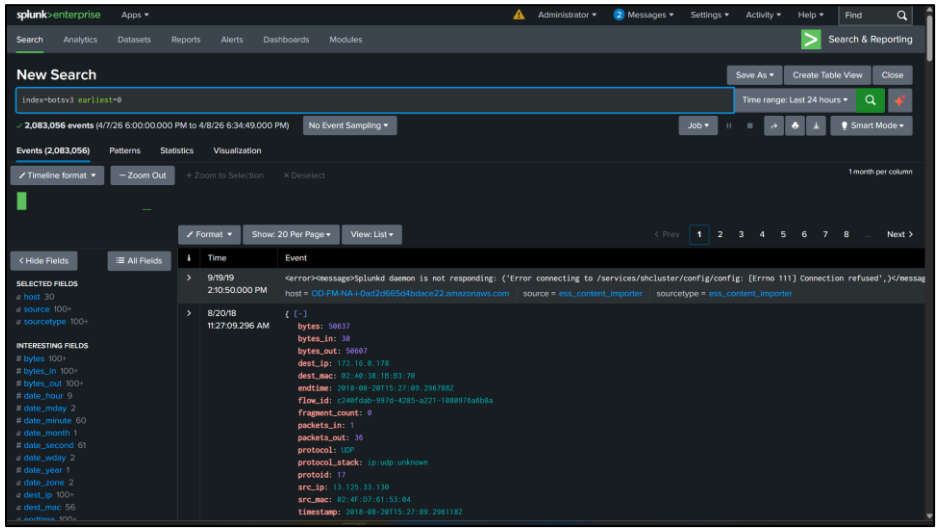


Figure 1 Ingested the BOTS v3 dataset into Splunk and verified successful log ingestion by querying all available events.

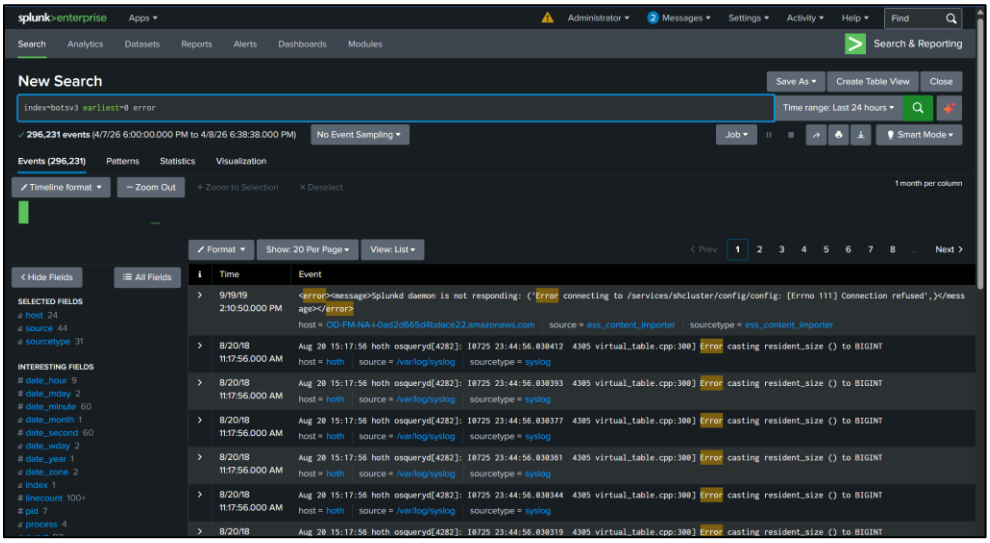


Figure 2 Filtered logs to identify system errors across the environment using keyword-based search.

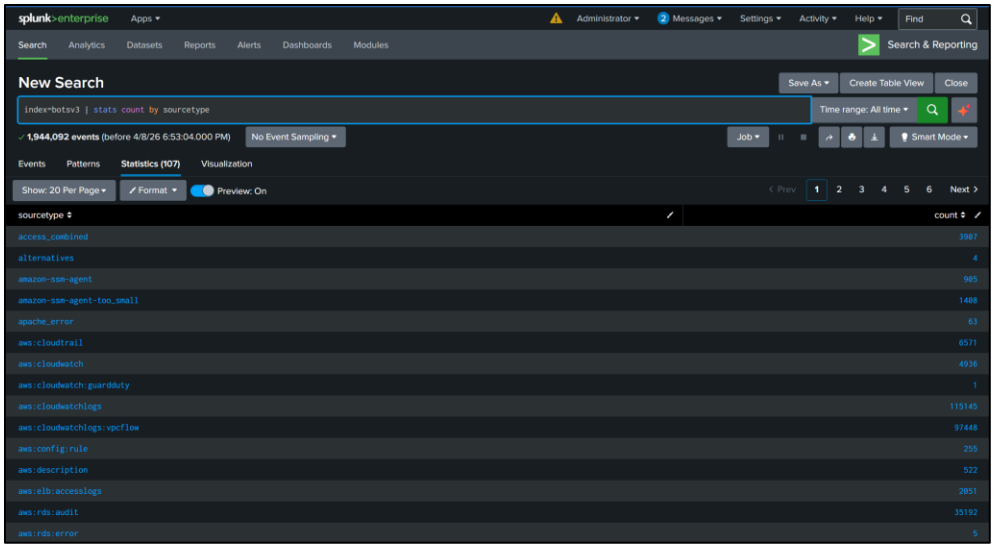
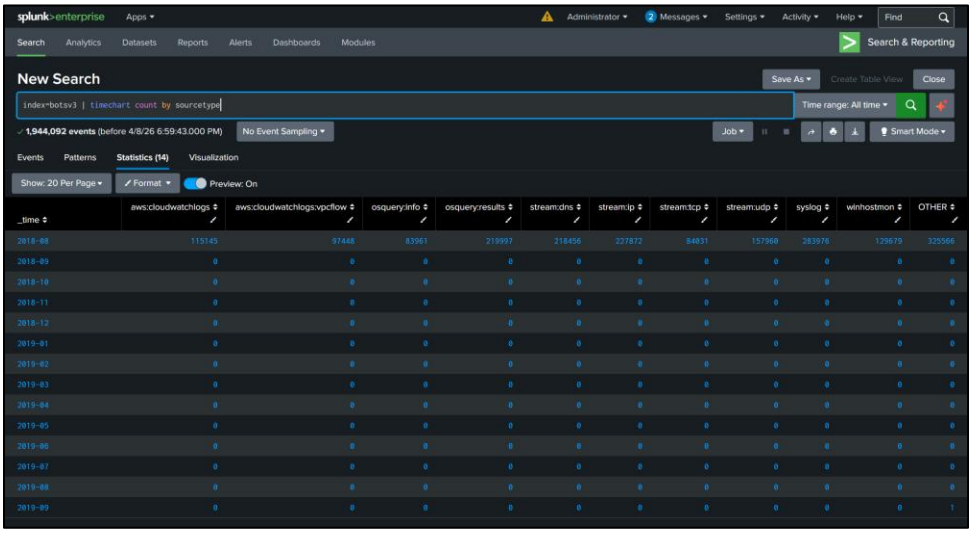


Figure 3 Analyzed log distribution by sourcetype to understand different data sources within the environment.



_time	aws:cloudwatchlogs	aws:cloudwatchlogs:vpcflow	osquery:info	osquery:results	stream:dns	stream:ip	stream:tcp	stream:udp	syslog	winhostmon	OTHER
2018-08	115145	97448	83961	219997	218456	227872	84831	157960	283976	129679	325566
2018-09	0	0	0	0	0	0	0	0	0	0	0
2018-10	0	0	0	0	0	0	0	0	0	0	0
2018-11	0	0	0	0	0	0	0	0	0	0	0
2018-12	0	0	0	0	0	0	0	0	0	0	0
2019-01	0	0	0	0	0	0	0	0	0	0	0
2019-02	0	0	0	0	0	0	0	0	0	0	0
2019-03	0	0	0	0	0	0	0	0	0	0	0
2019-04	0	0	0	0	0	0	0	0	0	0	0
2019-05	0	0	0	0	0	0	0	0	0	0	0
2019-06	0	0	0	0	0	0	0	0	0	0	0
2019-07	0	0	0	0	0	0	0	0	0	0	0
2019-08	0	0	0	0	0	0	0	0	0	0	0
2019-09	0	0	0	0	0	0	0	0	0	0	1

Figure 4 Visualized log activity over time to identify trends and anomalies across different data sources.

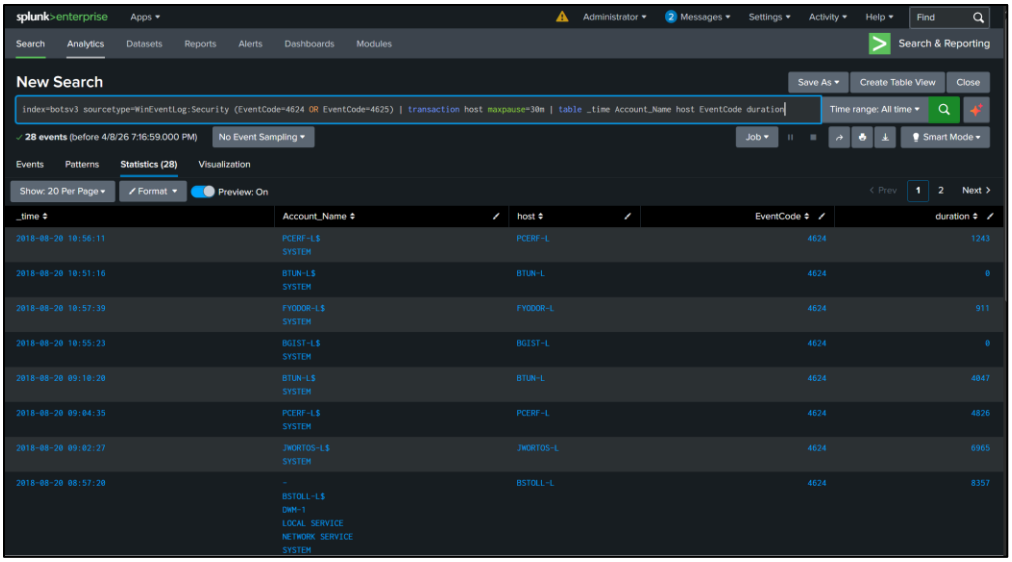


Figure 5 Identified failed login attempts (EventCode 4625) and analyzed frequency by user to detect potential brute-force attacks.

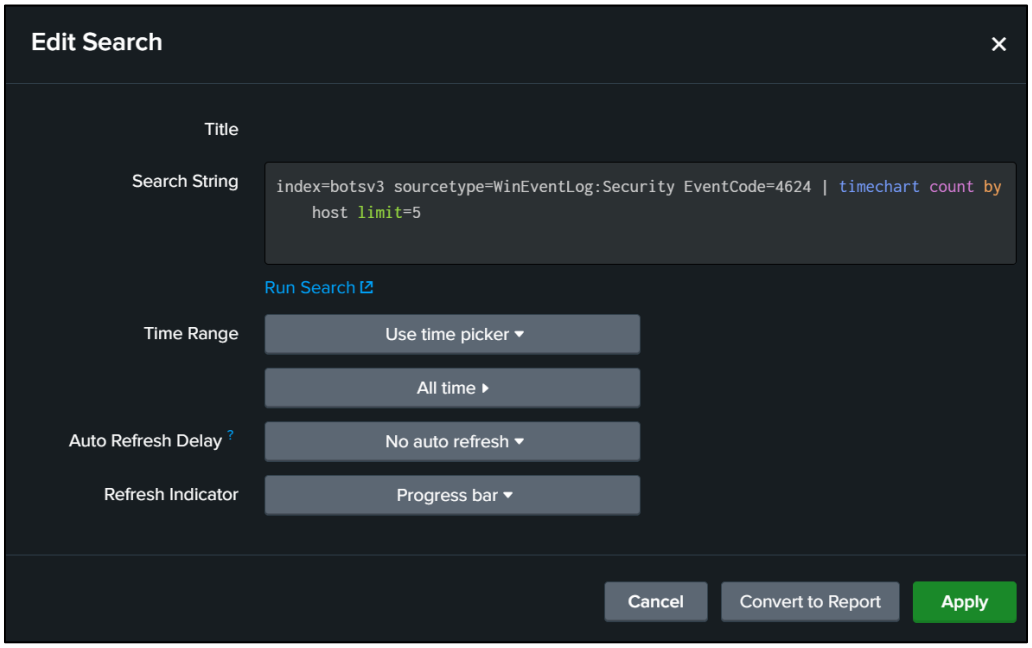


Figure 6 Created Splunk query to analyze successful login events (EventCode 4624) and visualize authentication activity across hosts.

Figure 7 Visualized authentication activity over time to identify trends and potential anomalies across hosts.

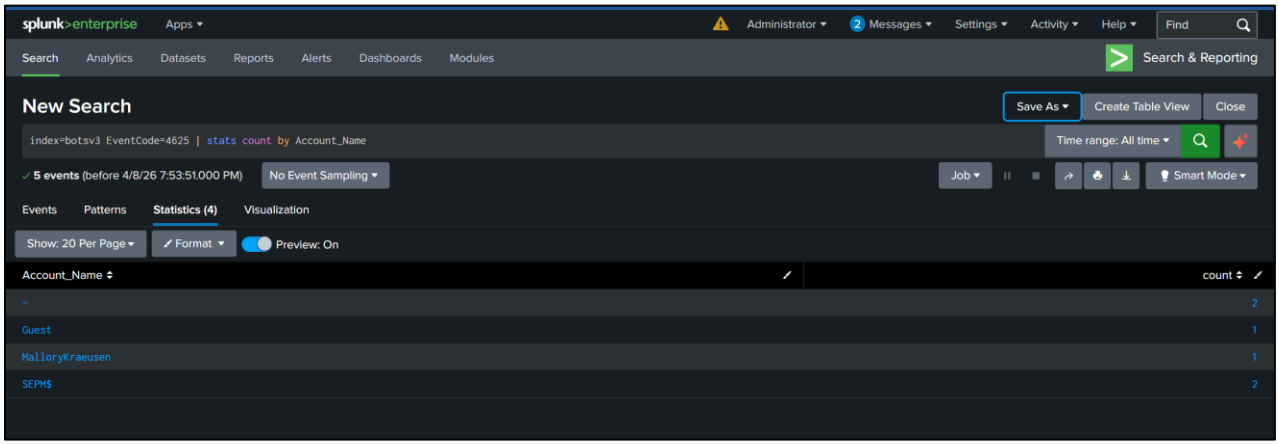


Figure 8 Analyzed failed login attempts (EventCode 4625) and grouped results by user to identify suspicious activity.

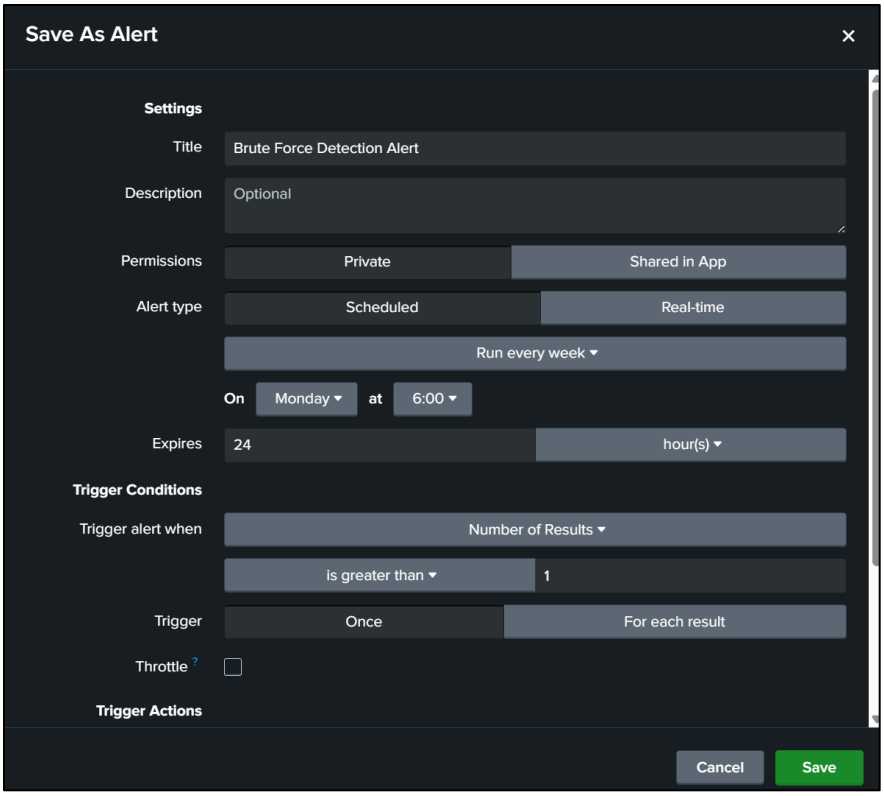


Figure 9 Configured a scheduled Splunk alert to detect potential brute-force login activity based on defined thresholds.

Figure 10 Created Splunk query to detect suspicious file modification activity by identifying high-frequency changes per user and file path.

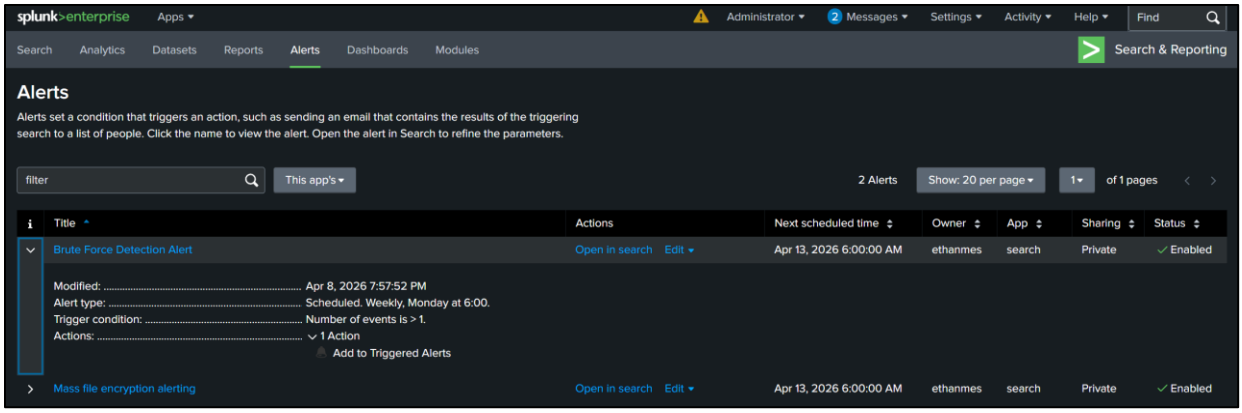


Figure 11 Configured and reviewed a brute-force detection alert to monitor suspicious login activity.

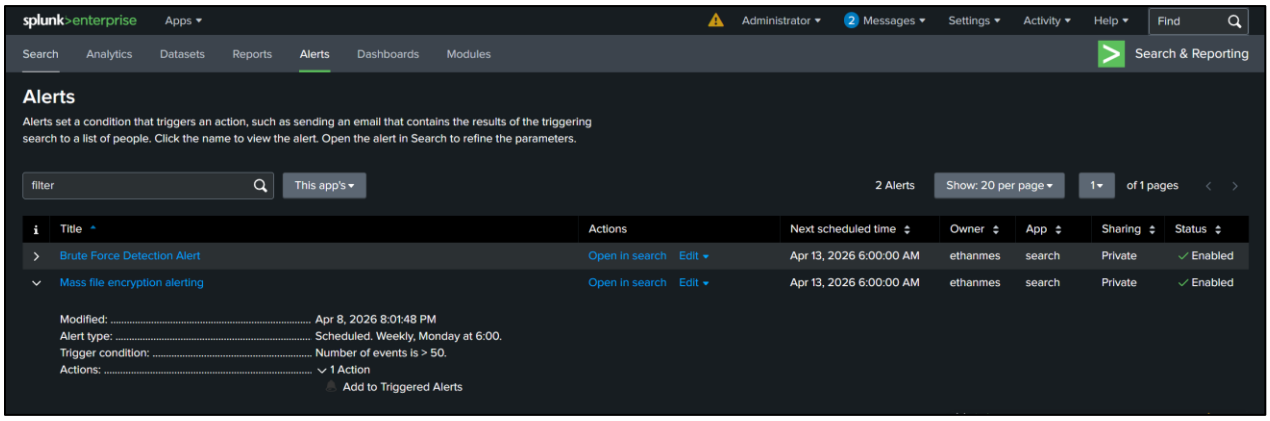


Figure 12 Created and validated a file modification alert to detect potential mass file changes or ransomware behavior.